Comprehensive Notes on Human-Computer Interaction

Based on Basic Design Rules & Research Methods (Chapters 2 & 3)

Detailed Compilation

Fall 2025 - Week 8

Contents

1 Part I: Basic Design Rules

This section covers the fundamental principles that influence how a design is perceived and utilized by users.

1.1 1. Accessibility

Accessibility refers to the practice of making designs usable by people of diverse abilities without the need for special adaptation or modification. It is often synonymous with *barrier-free design*, *universal design*, and *inclusive design*.

1.1.1 Four Characteristics of Accessible Design

1. **Perceptibility:** The design must be perceivable by everyone, regardless of sensory abilities (vision, hearing, touch).
2. **Operability:** Everyone must be able to use the design components (buttons, navigation) regardless of physical abilities.
3. **Simplicity:** The design should be easy to understand and use regardless of the user's experience, literacy, or concentration level.
4. **Forgiveness:** The design should minimize the occurrence and consequences of errors (e.g., providing "Undo" options).

Real-World Application: Assistive Technologies

- **Visual Impairment:** Screen readers (like NVDA or VoiceOver) require proper HTML tags (ARIA labels) to "perceive" a website.
- **Motor Impairment:** Elevators with buttons placed at a height reachable by wheelchair users (Operability).
- **Aural Feedback:** ATMs that provide headphone jacks for voice guidance.

1.2 2. The 80/20 Rule (Pareto Principle)

Also known as *Juran's Principle* or the *Vital Few and Trivial Many Rule*. It states that approximately **80% of effects are caused by 20% of the variables**.

1.2.1 Application in HCI

- **Usage:** 80% of a product's usage involves only 20% of its features.
- **Revenue:** 80% of company revenue often comes from 20% of products.
- **Errors:** 80% of system errors are usually caused by 20% of the bugs.

Design Strategy: Designers should focus resources on the critical 20% of features. Non-critical functions (the "trivial many") should be minimized or hidden to reduce clutter.

Real-World Application: Microsoft Office Ribbon

Microsoft analyzed user data and found that most users only used a small fraction of Word's capabilities (Bold, Italic, Paste, Save). These features were made prominent in the "Home" ribbon, while complex formatting tools were tucked away in other tabs.

1.3 3. Affordance

Affordance is a property where the physical characteristics of an object influence its function. It is a concept popularized by Don Norman.

- **Positive Affordance:** When the design implies the correct usage (e.g., a handle implies pulling).
- **Negative Affordance:** When the design implies an incorrect usage or prevents usage.

1.3.1 Mental Models

To create effective affordances, designs must match the user's **Mental Model** (what the user believes about how the system works).

- Hide implementation details (the code/backend) from the user.
- Make products simple, familiar, and flexible.
- Provide immediate feedback.

Real-World Application: Digital vs. Physical Affordance

- **Physical:** A flat plate on a door affords "pushing." A vertical handle affords "pulling." (The classic "Norman Door" problem occurs when a handle is placed on a push door).
- **Digital:** A button with a drop-shadow looks 3D and "pressable." A blue, underlined text universally affords a "hyperlink."

1.4 4. Hick's Law

Hick's Law describes the time it takes for a person to make a decision as a result of the possible choices he or she has.

> *The time required to make a decision increases logarithmically as the number of alternatives increases.*

1.4.1 The Equation

$$RT = a + b \log_2(n)$$

Where:

- RT = Response Time.

- a = Total time not involved in decision making.
- b = Empirical constant (processing time per option, approx 0.155s for humans).
- n = Number of equally probable alternatives.

Example Calculation: If detecting an alarm takes 2 seconds ($a = 2$) and there are 5 buttons ($n = 5$):

$$RT = 2 + 0.155 \times \log_2(5) = 2 + 0.155 \times 2.32 \approx 2.36 \text{ seconds.}$$

Real-World Application: Simplifying Menus

To reduce decision time, streaming services like Netflix categorize content into "Trending," "Action," etc., rather than showing an alphabetical list of 5,000 movies. This reduces n in the immediate view, lowering RT.

1.5 5. Fitts' Law

Fitts' Law predicts that the time required to rapidly move to a target area is a function of the ratio between the distance to the target and the width of the target.

1.5.1 The Equation

$$MT = a + b \log_2 \left(\frac{D}{S} + 1 \right)$$

Where:

- MT = Movement Time.
- a = Start/stop time (intercept, e.g., 0.230 sec).
- b = Speed of the device (slope, e.g., 0.166 sec).
- D = Distance from starting point to target.
- S = Size (width) of the target.

Key Takeaway: To decrease movement time, you must decrease distance (D) or increase target size (S).

Real-World Application: MacOS Dock & Edges

- **Infinite Width:** The edges of a screen (corners) are the easiest targets to hit because the cursor cannot overshoot them. This is why the Windows "Start" button and Mac "Apple" menu are in corners.
- **Context Menus:** Right-clicking opens a menu *at the cursor location*, reducing D to near zero.

1.6 6. The Golden Ratio & Fibonacci Sequence

The Golden Ratio (ϕ): Approximately **1.618**. It is found by dividing a line into two parts so that the longer part divided by the smaller part is also equal to the whole length divided by the longer part.

$$\frac{a+b}{a} = \frac{a}{b} = \phi \approx 1.618$$

Fibonacci Sequence: A series where a number is the sum of the two preceding ones $(0, 1, 1, 2, 3, 5, 8, 13...)$. The ratio of consecutive Fibonacci numbers approximates the Golden Ratio.

Real-World Application: Visual Harmony

Designers use the Golden Spiral to guide the eye.

- **Twitter Logo:** Constructed using circles with diameters corresponding to the Golden Ratio.
- **Typography:** Heading font sizes often follow the Golden Ratio relative to body text (e.g., 16px body $\times$ 1.618 $\approx$ 26px header).

1.7 7. Rule of Thirds

A composition technique where an image is divided into nine equal parts by two equally spaced horizontal lines and two equally spaced vertical lines. Important compositional elements should be placed along these lines or their intersections.

Real-World Application: Photography and Web Layouts

- **Ali vs. Liston:** The famous boxing photo shows Ali framed perfectly along the vertical third line, creating a dynamic composition.
- **Websites:** Hero images often place the main subject to the right or left third, leaving the other two-thirds open for text overlay.

1.8 8. Signal-to-Noise Ratio (SNR)

In design, this is the ratio of relevant information (Signal) to irrelevant information (Noise).

- **Signal:** The message you want to communicate.
- **Noise:** Visual clutter, unclear graphs, ambiguous icons, unnecessary decoration.

Goal: Maximize Signal, Minimize Noise.

Deep Dive: Data-Ink Ratio

Edward Tufte's concept of "Data-Ink Ratio" supports this. He argues that in a chart, any ink that does not represent data (like heavy grid lines, 3D effects on bar charts, or background patterns) is noise and should be removed.

2 Part II: Research Methods in HCI

This section details how to scientifically evaluate HCI systems using experimental research (Chapters 2 & 3).

2.1 1. Types of Behavioral Research

1. **Descriptive Investigations:** Focus on constructing an accurate description of *what* is happening.
   - *Methods:* Observations, field studies, focus groups.
   - *Claim:* "X is happening."
2. **Relational Investigations:** Identify relationships between multiple factors.
   - *Methods:* Correlations, surveys.
   - *Claim:* "X is related to Y." (Note: Correlation does not imply causation).
3. **Experimental Research:** Allows the establishment of a **causal relationship**.
   - *Methods:* Controlled experiments.
   - *Claim:* "X is responsible for Y."

2.2 2. Research Hypotheses

An experiment starts with a hypothesis—a precise problem statement testable through empirical investigation.

2.2.1 Types of Hypotheses

- **Null Hypothesis (H_0):** States there is **no difference** between experimental treatments. (e.g., "Mouse speed is equal to Touchscreen speed").
- **Alternative Hypothesis (H_1):** Mutually exclusive to H_0. States there **is a difference**. (e.g., "Mouse speed is NOT equal to Touchscreen speed").

Goal: The goal of an experiment is to find statistical evidence to *reject* the Null Hypothesis.

2.3 3. Experimental Variables

- **Independent Variable (IV):** The "Cause." The factor the researcher manipulates or controls.
  - *HCI Examples:* Device type (Mouse vs. Trackpad), Interface color, User Age, Context of use.
- **Dependent Variable (DV):** The "Effect." The outcome the researcher measures.
  - *HCI Examples:* Efficiency (time), Accuracy (error rate), Satisfaction (Likert scale), Cognitive load (NASA-TLX).

2.4 4. Randomization

To ensure validity, participants must be assigned to conditions randomly. This prevents bias.

- In a fully randomized experiment, neither the participant nor the investigator can predict the assignment.
- *Methods:* Random number tables, software-driven randomization (e.g., Python script, Excel).

2.5 5. Significance Tests and Errors

Significance tests determine if the results from a sample can be generalized to the entire population.

2.5.1 Type I and Type II Errors

Reality	**Decision: No Difference**	**Decision: Difference Exists**
Null is True (No Diff)	Correct	**Type I Error** (α)
Null is False (Diff Exists)	**Type II Error** (β)	Correct (Power)

Table 1: Error Types Matrix

- **Type I Error (α):** False Positive. "Gullibility." Rejecting the null hypothesis when it is actually true.
- **Type II Error (β):** False Negative. "Blindness." Failing to reject the null hypothesis when it is false.

Controlling Errors:

- **Alpha (α):** Usually set to 0.05 (5% chance of Type I error).
- **Statistical Power ($1 - \beta$):** The probability of successfully rejecting a false null hypothesis.
- Trade-off: Lowering α reduces Type I errors but increases the risk of Type II errors. In research, Type I errors are generally considered worse (claiming a discovery that doesn't exist).

2.6 6. Experimental Design Structures

2.6.1 A. Between-Group Design (Between-Subject)

Each participant experiences **only one** condition.

- **Advantages:** No learning effect (cleaner data), shorter time per user.
- **Disadvantages:** Individual differences (one group might just be smarter/faster naturally), requires larger sample size.
- **Use When:** Tasks are simple, or learning effects would ruin the study (e.g., solving a puzzle).

2.6.2 B. Within-Group Design (Within-Subject)

Each participant experiences **all** conditions.

- **Advantages:** Requires fewer participants, controls for individual differences (comparing a user against themselves).
- **Disadvantages: Learning Effect** (users get better at the task over time), fatigue.
- **Use When:** Participant pool is small, or individual variance is high.

2.6.3 C. Factorial Design

Used when investigating more than one Independent Variable.

- Allows the study of **Interaction Effects**.
- **Interaction Effect:** When the effect of one variable depends on the level of another.
- *Example:* A touchscreen might be faster than a mouse for *novices*, but slower than a mouse for *experts*. The effect of "Device" depends on "Experience."

2.7 7. Reliability and Systematic Errors

- **Random Errors (Noise):** Occur by chance. Can be reduced by increasing sample size.
- **Systematic Errors (Bias):** Push values in one specific direction. Cannot be fixed by sample size.
- **Sources of Bias:**
  1. Measurement instruments (e.g., a stopwatch running fast).
  2. Experimental procedures (e.g., testing one group in the morning and one at night).
  3. Participants (e.g., recruiting only tech-savvy students).
  4. Experimenter behavior (subconsciously guiding participants).
  5. Environment (noise, temperature).

2.8 8. Lifecycle of an Experiment

1. **Identify Hypothesis:** Define the problem.
2. **Specify Design:** Choose Between-subject vs Within-subject.
3. **Pilot Study:** A "dry run" to test the procedure and equipment.
4. **Recruit Participants:** Ensure they match target demographics.
5. **Run Data Collection:**
   - Greet & Consent.

- Training tasks.
- Actual tasks.
- Debriefing & Payment.

6. **Analyze Data:** Run statistical tests (T-test, ANOVA).
7. **Report Results:** Publish findings.